Lesson 7.6 Guided Notes

Session 1 · Distance Between Points on a Grid

I can find the distance between two points that share a coordinate. · 6.NS.C.8

Word / Spanish	Meaning — complete the blank	Example
distance / distancia	How far apart two points are. It is never _____.	(2, 1) to (2, 5) is 4 units
absolute value / valor absoluto	How far a number is from _____, ignoring its sign.	|−7| = 7
horizontal distance / distancia horizontal	Distance side to side. Use the _____-coordinates.	(−3, 2) to (4, 2)
vertical distance / distancia vertical	Distance up and down. Use the _____-coordinates.	(5, −1) to (5, 6)

Key idea

If two points share a y-coordinate, find the distance with the _____-coordinates.

If the changing coordinates have opposite signs, _____ their absolute values. If they have the same sign, _____ them.

Worked example

Find the distance from (−4, 3) to (5, 3).

The points share the y-coordinate _____, so use the x-coordinates.

−4 and 5 have _____ signs, so the trip crosses the y-axis.

Add the absolute values: 4 + 5 = _____.

Answer: The distance is _____ units.

You try

1. Find the distance between each pair of points.
a. (1, 6) and (1, −3) _____
b. (−7, −2) and (−2, −2) _____

_____

2. Point A is at (3, −1). Point B is 5 units directly above A. Name point B.

_____

Sum it up: To find the distance between two points that share a coordinate, I ___ because ___.

_____

Session 2 · Comparing Distances on the Plane

I can decide whether to add or subtract to find a distance, then compare two distances. · 6.NS.C.8

Word / Spanish	Meaning — complete the blank	Example
quadrant / cuadrante	One of the four _____ of the coordinate plane.	(−3, −5) is in Quadrant III
compare / comparar	Decide which amount is _____ or less.	12 units > 9 units
scale / escala	How much real length one grid _____ stands for.	1 unit = 5 meters

Key idea

Find _____ distance first. Then compare the two numbers.

With a scale, change _____ distances to real units before you compare.

Worked example

A = (1, 2), B = (1, 9), C = (−8, 2). Each unit is 5 meters. Is B or C farther from A?

A to B: same x, so 9 − 2 = _____ units = 35 m.

A to C: same y, opposite signs, so 1 + 8 = _____ units = _____ m.

Answer: _____ is farther, by 10 m.

You try

1. Add or subtract? Find each distance.
a. (−2, 3) to (7, 3) _____
b. (5, 3) to (12, 3) _____

_____

2. From (4, −3), Ana walks to (4, 6) and Ben walks to (−6, −3). Who walks farther?

_____

Sum it up: To compare two distances on the plane, I ___ because ___.

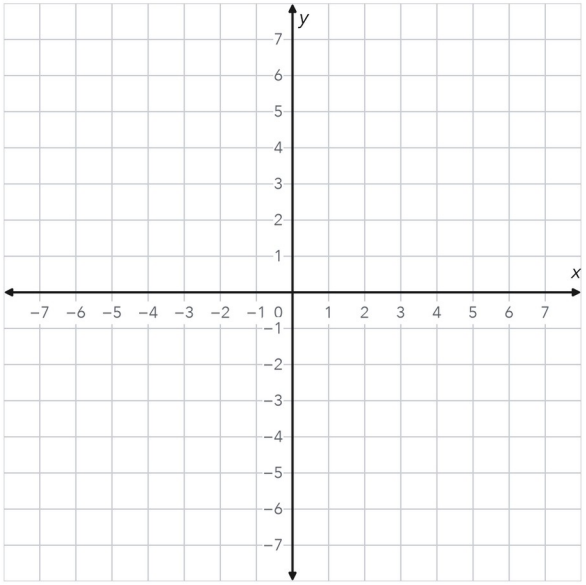
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Session 1 · Model, reason, and explain

Distance Between Points on a Grid · 6.NS.C.8

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Find distance from (−3,2) to (4,2), and from (−3,2) to (−1,2). Explain the shared coordinate and why the distances differ.



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_____

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Extend your reasoning

Create a contrasting example that exposes the misconception: uses the unchanged coordinate or returns a signed distance.. Explain what changes and what stays true.

_____

_____

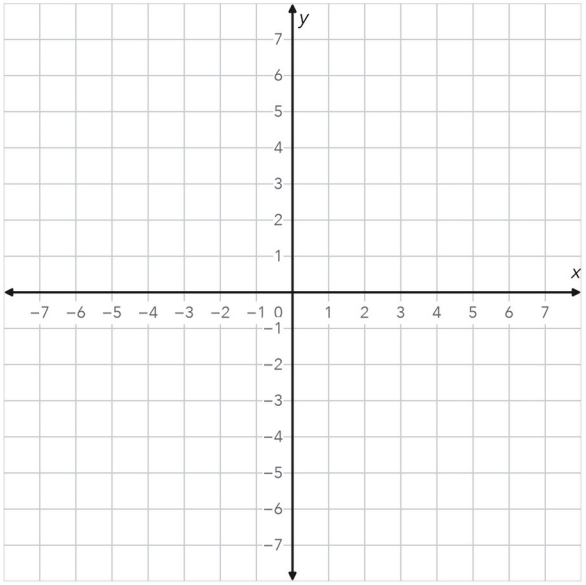
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Comparing Distances on the Plane · 6.NS.C.8

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A=(1,2), B=(1,8), C=(−7,2). Each grid unit is 3 m. Compare AB and AC in meters and state which is longer.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: compares grid units with meters without conversion.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?